

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4619: Peripherals and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What is an Embedded System? How does it differ from typical Computer Systems? 3 + 7
(CO1)
(PO1)
 - b) Draw the block-diagram of an embedded system considering a designer's view. Also, write down the customer functions for an embedded system. 4 + 4
(CO2)
(PO2)
 - c) Draw an optimized block diagram of a PCB considering a set of peripherals with two-set 4-LEDs, two-set 7-Segment Displays and one-set 8 × 8 Dot-Matrix Display connected to a single CPU using multiple IC 8255 PPIs. 7
(CO4)
(PO3)
2. a) Explain the importance of a priority resolver IC. Differentiate between Daisy-Chain and Multi-level Bus architectures with proper explanation. 3 + 7
(CO2)
(PO2)
 - b) Make the lists of the operating modes and operation command words of 8259 PIC. Using a block diagram, explain the functions of different registers of 8259 PIC. 4 + 4
(CO2)
(PO1)
 - c) Derive all data bits for Port-A, Port-B and Port-C of IC 8255 considering the (4, 5) and (5, 4) LEDs selected with green colors in an 8 × 8 Dot-Matrix Display. You can assume any kind of logic bit (i.e., 0 or 1) for unique selections for port data. 7
(CO4)
(PO2)
3. a) Explain why Programmed I/O is not suitable for Disk RD/WR. Listing different applications for Disk RD/WR, explain a suitable solution in this regard. 5 + 5
(CO2)
(PO1)
 - b) Using an appropriate block diagram, mention how the IC 8237 DMA Controller uses the system buses. 8
(CO2)
(PO1)
 - c) Define UART and USART. Draw the interfacing structure of IC 8251. 4 + 3
(CO2)
(PO1)
4. a) Explain Wired-AND concept. CAN bus protocol removes a complete graphs connection complexity for an embedded system Justify the statement with proper explanation. 3 + 7
(CO3)
(PO1)
 - b) Write short notes on following frames type of CAN bus: 4 + 4
(CO3)
(PO1)
 - i. Remote frame
 - ii. Error frame

- c) Suppose, in a CAN bus protocol, four bytes of data (10101001, 11110000, 00111011, and 00010001) from one IC are consecutively transmitted to another IC. Considering the number of bits, $n = 8$ for Check-sum approach, show the operations for error-free data transfer between the ICs. 7
(CO3)
(PO3)
5. a) Explain by Arbitration in BUS protocols. "Multi-master concept exists for I²C bus" — verify the statement with proper explanation. 3 + 7
(CO3)
(PO1)
- b) Explain different types of acknowledgement and negative-acknowledgement transmissions in I²C bus with appropriate signal diagrams. 8
(CO3)
(PO1)
- c) Draw the data formats of I²C protocol when the Master IC reads and writes to/from Slave IC. 7
(CO3)
(PO3)
6. a) Define Jitter? Explain the concepts of Isochronous interfacing with appropriate examples. 3 + 7
(CO1)
(PO1)
- b) Write short notes on the following wireless interfacing concepts with their comparison table: 5 × 3
(CO3)
(PO1)
- IrDA
 - Bluetooth
 - IEEE 802.11
 - IEEE 802.15.4
 - LoRaWAN